

From Lightweight Ceramic HEAT-SINKS To INTERFACE SOLUTIONS For Heat Dissipation



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While going through electronics product reviews, you might have come across phrases like, "The phone heats up on intensive gaming" or "The device feels warm after constant use." According to Ramanan, vice president - engineering, defence solutions, Mistral Solutions, "Inefficiency of electronics products leads to heat dissipation, which results in thermal analysis requirement."

Electronics products generate heat on usage, most of which is by switching devices like metal oxide semiconductor field effect transistors and integrated circuits. This heat has to be removed from the device for efficient functioning.

Thermal solutions come in handy while thinking of a way to solve this problem. Heat dissipation needs of today's components are more challenging than ever. System airflow and its impact on heat-sink design is a very important factor in applications generating sufficient heat. "Telecom, automobiles and lighting are some applica-

tions that require major use of thermal solutions," says Dr Misra, director - research and development, Henkel Adhesive Technologies.

Consider an efficient heat-sink

Ongoing increase in power usage among devices such as processors and insulated-gate bipolar transistors results in higher capacity cooling requirements to remove excess heat. To cool hotter components, you may use large fans, heat-sinks and increase surface area. Cooling performance can also be improved by using a higher-performance interface material between the case and the heat-sink.

Heat transfer occurs when two surfaces have different temperatures, thus causing heat energy to transfer from the hotter surface to the colder one. Some tips from industry experts in the box titled 'Application tips for heat-sink design,' should be helpful in designing one.

The latest in heat removal

"The amount of heat generated and the rate of generation are essential to arrive at the size and design of the heat-sink," says Ramanan. "We tend to choose the thermal pad based on these two factors," he adds. On a more general note, radial and grid patterns (pin fin) are being used by designers today for better heat dissipation. Radial fins provide ease in design, while grid patterns allow for a much higher surface area to aid heat loss.

As for the material, aluminium has been the metal of choice for the past decade for efficient cooling. Aluminium also remains popular for its conductivity, light weight and low cost, relative to other metals such as copper. To create more complex designs, manufacturers have also employed high-quality alloys like forged aluminium.

Ceramics, thermally-conductive plastics and graphite are also being experi-

Fig. 1: Applying thermal paste on the processor

